LAB 3: ARCHITECTURE SELECTION & JUSTIFICATION

Pharmacy Expiry & Re-order Dispatch Engine

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Architecture Selection

We chose **Layered Architecture** for the Pharmacy Expiry & Re-order Dispatch Engine.

Architectural Choice

The system is divided into three clear layers: Presentation, Business, and Data. The Pharmacy Management UI forms the Presentation layer. Inventory Management, FEFO Dispensing, and Stock Monitoring & Re-order handle the main business logic. The Inventory Database belongs to the Data layer.

Reason 1 – Clear Separation of Pharmacy Functions

The system has different responsibilities such as tracking batch expiry, preparing FEFO dispensing lists, monitoring stock levels, and generating purchase orders. Keeping these functions in the Business layer separates them from the user interface and database, making the system easier to understand and maintain.

Reason 2 – Reliable Inventory Processing

FEFO and safety-stock rules must be applied consistently because incorrect inventory processing can result in the wrong medicine batch being selected or a required order being missed. A separate Business layer keeps these rules together while the Data layer handles the storage of inventory and batch information.

Security Advantage

The UI does not access the database directly. Requests pass through the Business layer, where validation and access checks can be applied before inventory or purchase order data is changed. This provides better control over sensitive pharmacy data.

Performance Benefit

The layers can be optimized separately. For example, database queries for stock and batch information can be optimized without changing the UI or the FEFO and re-order logic. This helps the system respond efficiently as the number of inventory operations increases.